

Boundary Layer Equations

Velocity B.L.: $\frac{\partial^2 u}{\partial x^2} \ll \frac{\partial^2 u}{\partial y^2}$

Thermal B.L.: $\frac{\partial^2 T}{\partial x^2} \ll \frac{\partial^2 T}{\partial y^2}$

Cons. Mass: $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$

Newton's 2nd Law of Motion:

$$\frac{u \partial u}{\partial x} + \frac{v \partial v}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \frac{\partial^2 u}{\partial y^2}$$

Net rate x-momentum leaves the control volume.

Net Pressure force

Net force due to viscous shear stress

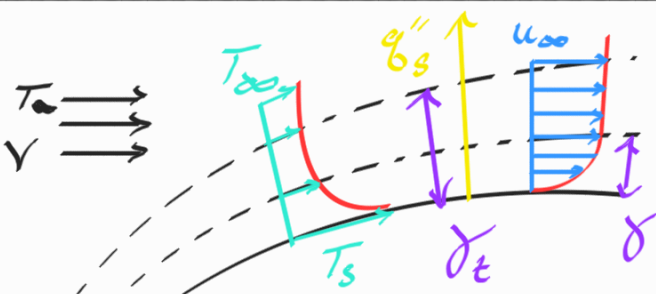
Cons. Energy:

$$\frac{u \partial T}{\partial x} + \frac{v \partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2} + \frac{\nu}{C_p} \left(\frac{\partial u}{\partial y} \right)^2$$

Net thermal energy leaves the control volume due to fluid motion

inflow of thermal energy due to convection in y direction

Remaining viscous dissipation



Boundary Layer Similarity

- * $u = f(x, y, L, V, \mu, \rho)$
- * $T_s = f(x, L, V, \mu, \rho, k)$
- * $T = f(x, y, L, V, \mu, \rho, k, T_s, T_\infty)$
- * $h = f(x, L, V, \mu, \rho, k, T_s, T_\infty)$

Neglect viscous dissipation gives Normalized forms of x-moment. and energy equations

$$\frac{u^* \partial u^*}{\partial x^*} + \frac{v^* \partial u^*}{\partial y^*} = -\frac{\partial p^*}{\partial x^*} + \frac{1}{Re_L} \frac{\partial^2 u^*}{\partial y^{*2}}$$

$$\frac{u^* \partial T^*}{\partial x^*} + \frac{v^* \partial T^*}{\partial y^*} = \frac{1}{Re_L Pr} \frac{\partial^2 T^*}{\partial y^{*2}}$$

$$Re_L = \frac{\rho V L}{\mu} = \frac{VL}{\nu} \rightarrow \text{Reynolds \#}$$

$$Pr = \frac{c_p \mu}{k} = \frac{\nu}{\alpha} \rightarrow \text{Prandtl \#}$$

$$\frac{\delta}{\delta_t} \propto Pr^{1/4}$$

gases $Pr \sim 1$ Liquid Metals $Pr \ll 1$ Oils $Pr \gg 1$

Dimensionless Local Convection Coef. (Nusselt #)

$$Nu_x = \frac{h x}{k_s} = f(x^*, Re_x, Pr) = \frac{\text{Convection (fluid)}}{\text{Conduction (solid)}}$$

$$Nu_L = \frac{h L}{k_s} = f(Re_L, Pr)$$

* Turbulence Occurs @ Crit Re^* i.e. $Re_{crit} \approx 2300$ i.e.

- Hydrodynamic Entry Length: Laminar $\frac{x_{fd}}{L} = 0.05 Re_D$ Turbulent $10 < \frac{x_{fd}}{L} < 60$

- Thermal Entry Length: Laminar $\frac{x_{fd}}{L} = 0.05 Re_D Pr$ Turbulent $10 < \frac{x_{fd}}{L} < 60$

* Fully Developed Conditions (Assuming: - Steady flow - Constant Prop. (W, h constant))

- Friction Factor: $f = \frac{(\Delta P) D}{L \rho U_m^2}$ or $f = \frac{2 \tau_w}{\rho U_m^2}$

- Laminar: $f = 64 / Re_D$

- Turbulent (smooth circular): $f = (0.790 \ln(Re_D) - 1.64)^{-2}$

- Turbulent (rough circ. tube) use table/graph * Relative Roughness = $\frac{e}{D}$

- Pressure Drop: $\Delta P = P_1 - P_2 = \frac{f L \rho U_m^2}{2 D} (X_2 - X_1)$

- Power Requirement: $P_{req} = \Delta P Q$ or $P_{req} = \frac{\Delta P \dot{V}}{\eta}$

- Requirement for Sol. Thermal Cond. $\frac{\partial}{\partial x} \left(\frac{T_s - T_\infty}{T_s - T_m} \right) = 0$

- Effect on local Conv. Coef. $\frac{\partial}{\partial x} \left(\frac{T_s - T_\infty}{T_s - T_m} \right) = \frac{\partial T_s}{\partial x} \frac{1}{T_s - T_m} - \frac{\partial T_m}{\partial x} \frac{1}{T_s - T_m} \neq f(x)$

Assuming Constant Prop. $\Rightarrow \frac{\delta_s}{\delta_t} = \frac{1}{Pr} \neq f(x) \Rightarrow h \neq f(x)$

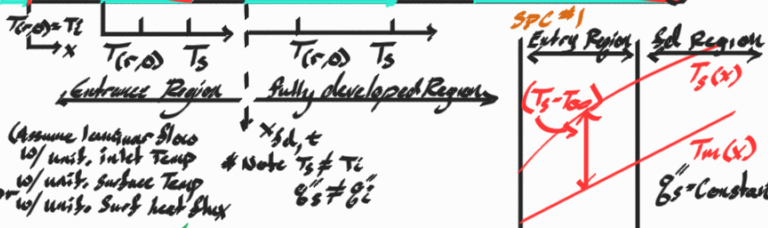
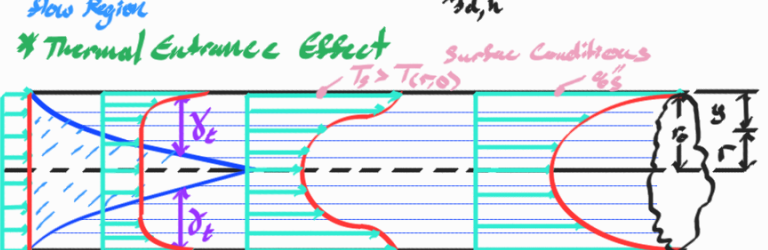
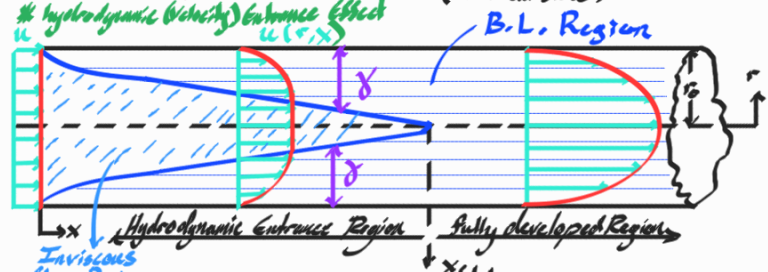
Variables:

- Size (L)
- Loc. (x, y)
- Velocity (V)
- Fluid Properties: ρ, μ, c_p, k
- Energy in y $\rightarrow$ u energy in x $\rightarrow$
- μ = Dynamic Viscosity
- ν = Kinematic Viscosity
- δ = Hydro. Dyn. B.L. thickness
- δ_t = Thermal B.L. thickness

D.L. Variables:

- $x^* = x/L$
- $y^* = y/L$
- $u^* = u/V$
- $v^* = v/V$
- $T^* = \frac{T - T_s}{T_\infty - T_s}$

Internal Pipe Flow (i.e.)



* Mean Velocity & Temp (U_m, T_m)

$$\dot{m} = \rho U_m A_c = \int_{A_c} \rho u(r, x) dA_c \Rightarrow U_m = \frac{\int_{A_c} \rho u(r, x) dA_c}{\rho A_c}$$

$$T_m = \frac{\int_{A_c} \rho u c_p T dA_c}{\int_{A_c} \rho u c_p dA_c} \text{ - Associated w/ flow thru cross sec.}$$

$$T_m = \frac{2}{U_m \rho c_p} \int_0^R u(r, x) T(r, x) r dr \text{ - for incompressible fluid with constant prop. in pipe.}$$

$$q_s'' = h_x (T_s - T_m) \text{ - for local heat flux}$$

* Hydrodynamic and Thermal Entry Lengths (Laminar or turbulent flow depends on Re)

$$Re_D = \frac{\rho U_m D}{\mu} \quad D_h = \frac{4 A_c}{P} \quad A_c = \text{Area C.S.} \quad P = \text{Perimeter}$$

for Circular pipe: $Re_D = \frac{4 \mu}{\pi \rho U_m D}$ or Diameter

* Using Mean Temp to Calc q_{conv}

$$q = \dot{m} C_p (T_{m,o} - T_{m,i}) = h A (T_s - T_m) \quad q = h A (T_s - T_{m,o})$$

- Special Case 1 (Unit. Heat Flux) $T_m(x) = T_{m,i} + \frac{q_s'' P x}{\dot{m} C_p}$ P = Perimeter

- Special Case 2 (Uniform Surf. Temp) $\Delta T \equiv T_s - T_m$

$$q_{conv} = q = h A_s \Delta T_{lm} \quad T_{lm} = \frac{\Delta T_o - \Delta T_i}{\ln(\frac{\Delta T_o}{\Delta T_i})}$$

$$\frac{\Delta T_o}{\Delta T_i} = e^{(\frac{P L}{\dot{m} C_p} h)} = e^{(\frac{P L}{\dot{m} C_p} h)}$$

$$\Delta T_o = T_s - T_{m,o} \quad \Delta T_i = T_s - T_{m,i}$$

$$q_{conv} = q = q_s'' A_s = q_s'' (\pi D L)$$

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* Finding h for SD Flow in Circular tubes

- Laminar (local Nusselt #)
- * Unif. Surf. heat flux (q_s''): $Nu_D = 4.36 = \frac{hD}{k}$
- * Unif. Surf. Temp (T_s): $Nu_D = 3.66$
- Turbulent
- * Smooth Surf + fully turbulent:

$$1^* Nu_D = 0.023 Re_D^{4/5} Pr^n \left\{ \begin{array}{l} n=0.3, T_s < T_m \\ n=0.4, T_s > T_m \end{array} \right.$$

x effects of wall roughness to transitional flow (μ constant or T_s is constant)

$$2^* Nu_D = \frac{(f/8)(Re_D - 1000)Pr}{1 + 12.7\sqrt{f/8}(Pr^{2/3} - 1)}$$

* Finding h for SD Flow in Non Circular tube

- use $D_h = \frac{4A_c}{P}$ (hydraulic diameter)
- laminar: use Table 8.1 for Nu_D | $\bar{h} = h$ (constant) for laminar flow
- dependent on Surf. conditions and duct Aspect ratio
- Turbulent: Use either 1* or 2* regardless the surface conditions.

Heat Exchangers

* Temp Profile by HX type

- Counter-Flow HX

$$\Delta T_1 = T_{h,1} - T_{c,1} \quad \Delta T_2 = T_{h,2} - T_{c,2}$$

$$= T_{h,i} - T_{c,o} = T_{h,o} - T_{c,i}$$

- Parallel HX

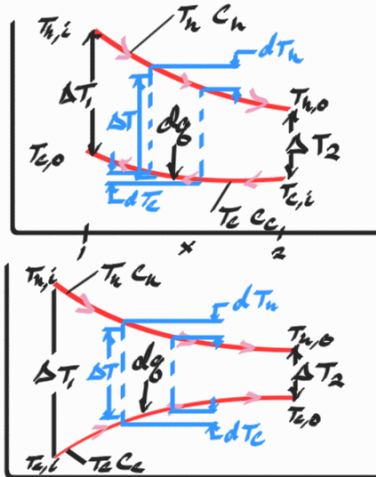
$$\Delta T_1 = T_{h,i} - T_{c,i} \quad \Delta T_2 = T_{h,o} - T_{c,o}$$

* CF HX transfers more heat

$$\Delta T_{lm,CF} > \Delta T_{lm,PFHX}$$

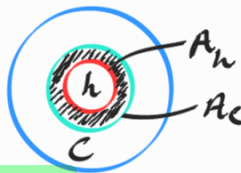
* Special Operating Cond.

- Case a) $C_h \gg C_c$ or h is a Cond Vapor $C_h \rightarrow \infty$
 - Negligible or no change in T_h
- Case b) $C_c \gg C_h$ or c is a evap Vapor $C_c \rightarrow \infty$
 - Negligible or no change in T_c
- Case c) $C_h = C_c$



* HX Analysis Determine (U) overall

- Wall separating 2 fluid streams (clean unfouled surf.)



$$UA^{-1} = (UA_c)^{-1} = (UA_h)^{-1} = R_w + (UA_c)^{-1} + (UA_h)^{-1}$$

- w/ fouling

$$UA^{-1} = (r_o h A_c)^{-1} + (r_o A_c)^{-1} R_{f,c} + R_w + (r_o A_h)^{-1} R_{f,h} + (r_o h A_h)^{-1}$$

$$R_w = \ln\left(\frac{r_o}{r_i}\right) / (2\pi L k_{wall})^{-1}$$

$$R_f = \text{fouling factor (* use Table 11.1)}$$

* HX Performance Analysis #1

(LMTD) (useful when outlet Temp known)

$$\dot{Q} = \dot{m} c_p \Delta T$$

- H.T. = Thermal Energy

$$\dot{Q} = \dot{m}_h (\dot{Q}_{h,i} - \dot{Q}_{h,o}) = \dot{m}_h c_p \Delta T_h$$

$$\dot{Q} = \dot{m}_c (\dot{Q}_{c,i} - \dot{Q}_{c,o}) = \dot{m}_c c_p \Delta T_c$$

$$\dot{Q} = UA \Delta T_{lm} \quad \left| \quad T_{lm} = \frac{\Delta T_2 - \Delta T_1}{\ln\left(\frac{\Delta T_2}{\Delta T_1}\right)} \right. \\ \text{or} \quad \left. = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)} \right.$$

* HX Analysis #2 ϵ -NTU

(useful when outlet Temps unknown)

- HX ϵ (effectiveness) (NTU $\rightarrow$ Table 11.3) $\epsilon \rightarrow$ Table 11.4

$$\epsilon = \frac{\dot{Q}}{\dot{Q}_{max}} \quad 0 \leq \epsilon \leq 1$$

$$\dot{Q}_{max} = C_{min} (T_{h,i} - T_{c,i})$$

$$C_{min} = C_h \text{ if } C_h < C_c$$

$$= C_c \text{ if } C_h > C_c$$

$$\dot{Q} = \dot{Q}_{max} = \epsilon C_{min} \Delta T_i$$

$$\epsilon = \frac{\dot{Q}_h}{\dot{Q}_{max}} = \frac{\dot{Q}_c}{\dot{Q}_{max}}$$

$$\epsilon = \frac{C_h \Delta T_h}{C_{min} \Delta T_i} = \frac{C_c \Delta T_c}{C_{min} \Delta T_i}$$

$$\epsilon = f(NTU, \frac{C_{min}}{C_{max}})$$

- NTU (# Transfer Units)

$$NTU = \frac{UA}{C_{min}}$$

* Design Calc.

$$NTU = f(\epsilon, Cr)$$

$$Cr \text{ (heat cap. ratio)} = \frac{C_{min}}{C_{max}}$$

for All HX $\epsilon \uparrow$ $Cr \downarrow$

- Special case $C_{max} \rightarrow \infty$ for (evap, boil, cond) for $Cr = 0$ $\epsilon = 1 - e^{-NTU}$ or $NTU = -\ln(1 - \epsilon)$

Modes of heat transfer

Conduction: $q'' = -k \nabla T$

Convection: $q'' = h(T_s - T_{\infty})$

Radiation: $q'' = \epsilon \sigma (T_s^4 - T_{sur}^4)$

Energy Balance

$$\dot{E}_{sys} = \dot{E}_{in} - \dot{E}_{out} + \dot{E}_{gen}$$

$$\Rightarrow \rho c_p V \frac{dT}{dt} = \dot{q}_{net} + \dot{E}_{gen}$$

The heat Equation

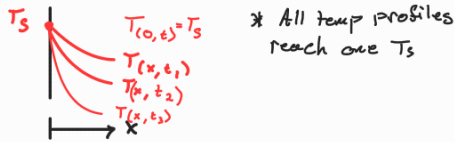
$$\rho c_p V \frac{dT}{dt} = \frac{\partial}{\partial x}(\rho k \frac{\partial T}{\partial x}) + \frac{\partial}{\partial y}(\rho k \frac{\partial T}{\partial y}) + \frac{\partial}{\partial z}(\rho k \frac{\partial T}{\partial z}) + \dot{q}$$

Boundary & Initial Conditions

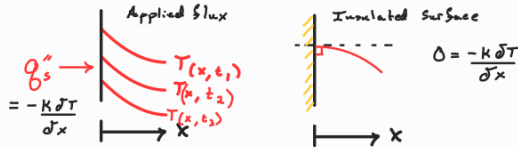
I.C $\rightarrow T(x, t) | t=0$

B.C.

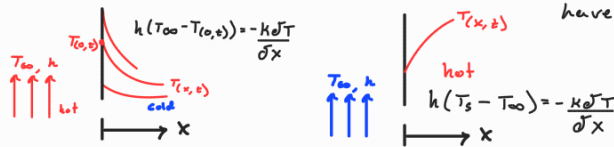
* Constant surface temp



* Constant heat flux



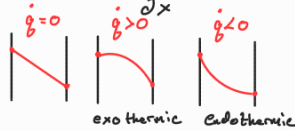
* Convection



* For S.S. 1-D Temp profiles

* w/ gen $\frac{d^2 T}{dx^2} = \frac{-\dot{q}}{k} \Rightarrow$ parabolic profile

* w/o gen $\frac{d^2 T}{dx^2} = 0 \Rightarrow$ linear profile



Nomenclature and Units

thermal energy	U (J) or U (J/m ³)
heat	Q (J)
heat rate	$\dot{Q}$ (W or J/s)
heat rate/length	$\dot{Q}'$ (W/m)
heat flux	$\dot{q}''$ (W/m ²)
heat generation	$\dot{q}$ (W/m ³)
emissivity	ϵ (DL)
Thermal diffusivity (absorptivity)	$\alpha = \frac{k}{\rho c_p}$

For Elec. Cir.

$$\dot{E}_{gen} = P = IV$$

Thermal Resistance and Thermal Circuits

Thermal Resistance: $R_t = \frac{\theta}{\dot{q}''}$

conduction plane wall: $R_{t,cond} = \frac{L}{kA}$

Convection: $R_t = \frac{1}{hA}$

Radiation: $R_t = \frac{1}{h_r A}$ | $h_r = \epsilon \sigma (T_s + T_{sur})(T_s^2 + T_{sur}^2)$

Thermal R for Unit A: $R_t = \frac{R_t''}{A}$ | Units: $R_t = \frac{K}{W}$ | $R_t'' = \frac{K \cdot m^2}{W}$ | $R_t = \frac{1}{hA}$ | $R_t'' = \frac{1}{h}$

Contact R: $R_{t,c} = \frac{T_{a,b} - T_{b,c}}{\dot{q}''}$ | $R_{t,c}'' = \frac{R_{t,c}}{A}$ | $R_{t,c}'' = \frac{1}{h_c}$ | $R_{t,c}'' = \frac{1}{h_c}$

Series: $R_{tot} = \sum R$

Parallel: $R_{tot} = (\sum R^{-1})^{-1}$

Special Cases

* 1-D cond, no $\dot{E}_{gen}$, Constant prop

$$\frac{\partial}{\partial x} \left(\frac{k \partial T}{\partial x} \right) = \rho c_p \frac{\partial T}{\partial t} \Rightarrow \frac{\partial^2 T}{\partial x^2} = \frac{\partial T}{\partial t}$$

$\rightarrow$ 2 B.C. + 1 I.C.

* S.S. 1-D cond. w/ generation

$$\frac{\partial}{\partial x} \left(\frac{k \partial T}{\partial x} \right) + \dot{q} = 0 \Rightarrow \frac{d^2 T}{dx^2} = \frac{-\dot{q}}{k}$$

$$\Rightarrow T(x) = \frac{-\dot{q} x^2}{2k} + C_1 x + C_2 \rightarrow 2 \text{ B.C.}$$

• Walls

$$T(x) = T_s + \frac{(T_{s2} - T_{s1})x}{L} \quad | \quad \dot{q}'', \dot{q}', \dot{q} \text{ constants}$$

• Cylinders

$$T(r) = \frac{T_{s1} - T_{s2}}{\ln(r/r_2)} \ln\left(\frac{r}{r_2}\right) + T_{s2} \quad | \quad \dot{q}', \dot{q} \text{ constant}$$

$\dot{q}''$ not constant

2-D systems

* For temp/adiabat profiles

adiabats $\perp$ intersect temp profiles

1-term Approximation * Use if $Bi < 0.1$

Determining Bi (Biot #)

$$Bi = \frac{h L_c}{k} \quad | \quad L_c = \frac{1}{2} \text{ plane wall's thickness}$$

$$= \frac{1}{2} \text{ long cylinders radius}$$

$$= \frac{1}{3} \text{ of spheres radius } L_c = \frac{V}{A_s} = \frac{4\pi r^3/3}{4\pi r^2} = \frac{r}{3}$$

$$L_c = L \text{ if insulated one side}$$

* equations given in equation sheet

• use table 5.1 for ξ_1 & C_1

* plane wall

$$\text{For Temp @ mid plane } x^* = 0 \rightarrow \theta_o^* = \frac{\theta_o}{\theta_i} = \frac{T_o - T_{\infty}}{T_i - T_{\infty}} \approx C_1 e^{-\xi_1^2 z_L^2}$$

$$\text{For Temp @ any } x \rightarrow \theta^* = \theta_o^* \cos(\xi_1 x^*)$$

$$\text{For heat transfer} \rightarrow \Delta E_{sys} = -Q \quad | \quad Q = \dot{Q}_0 \left(1 - \frac{\sin(\xi_1)}{\xi_1} \right) \theta_o^*$$

$\dot{Q}_0 = \rho c_p V \dot{\theta}_i$

Fin Arrays (See table 3.5)

$$A_b = (\# f) A_f + A_b$$

w/o contact Resistance

$$\dot{Q}_b = \# f \eta_f h A_f \theta_b + h A_b \theta_b \equiv \eta_o h A_b \theta_b = \frac{\dot{Q}_b}{R_{t,o}}$$

$$\eta_o = 1 - \frac{\# f A_f (1 - \eta_f)}{A_b}$$

$$R_{t,o} = \frac{A_b}{\dot{Q}_b} = \frac{1}{\eta_o h A_b}$$

w/ contact Resistance

$$\dot{Q}_b = \eta_{o,c} h A_b \theta_b = \frac{\dot{Q}_b}{R_{t,o,c}}$$

$$\eta_{o,c} = 1 - \frac{\# f A_f (1 - \eta_f)}{A_b} \quad | \quad C_1 = 1 + \eta_f h A_f \left(\frac{R_{t,o,c}^V}{R_{t,o,b}} \right)$$

$$R_{t,o,c} = \frac{1}{\eta_{o,c} h A_b}$$

Radiation

$$q''_{\text{net}} = q''_{\text{rad, emitted}} - q''_{\text{rad, absorbed}}$$

$$q''_{\text{net}} = \epsilon \sigma (T_s^4 - T_{\text{sur}}^4)$$

(+) out of solid

(-) into solid

radiation behaves like electromagnetic wave and can be characterised as

(1) wavelength, (2) frequency,

(3) speed rad. propagates through vacuum

$$c = 2.998 \times 10^8 \text{ m/s}$$

Flux (W/m^2)

Emissive Power, E

$$E = \epsilon \sigma T_s^4 \quad (\epsilon = \text{emissivity})$$

Irradiation, $G = q''_{\text{in}}$

Irradiation can be

- reflected (ρ reflectivity)

- absorbed (α absorptivity)

- transmitted (τ transmissivity)

radiosity, $J = q''_{\text{out}}$

For opaque surface

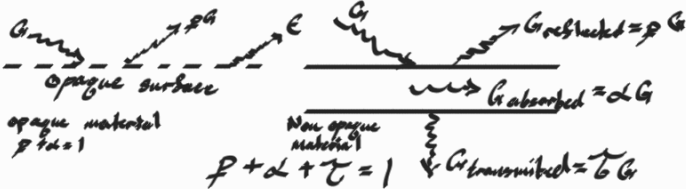
$$J = E + \rho G$$

Net radiative flux

For opaque surface

$$q''_{\text{rad}} = J - G$$

$$q''_{\text{rad}} = \epsilon \sigma T_s^4 - \alpha G$$



Black Body Radiation

- absorbs all radiation

- No surface can emit more energy than a B.B

- is a diffuse emitter/absorber

i.e. Perfect emitter $\epsilon = 1$

Perfect absorber $\alpha = 1$

- Emissive Power B.B. $E_b = \sigma T^4$ | $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \text{K}^4$

- Intensity B.B. $I_b = \frac{E_b}{\pi}$

- Surface Emissivity $\epsilon = \frac{E}{E_b}$

Relating q''_{rad} w B.B and G.B assumptions

$$q''_{\text{rad}} = \epsilon \sigma T_s^4 - \alpha G \rightarrow \text{for large surroundings}$$

$$= \epsilon \sigma T_s^4 - \alpha \sigma T_{\text{sur}}^4 \quad G = G_b = E_b = \sigma T_{\text{sur}}^4$$

$$= \epsilon \sigma (T_s^4 - T_{\text{sur}}^4)$$

Direction Considerations for I (intensity)

Spherical coordinates characterised by

- Zenith (or) polar angle θ

- Azimuthal angle ϕ

- amount of radiation emitted from surface (dA_s)

propagates in particular direction

is quantified by a differential solid angle

$$\text{Solid angle } d\omega = \frac{dA_n}{r^2}$$

$$I_{\lambda, e}(\lambda, \theta, \phi) = \frac{dq}{(dA_s \cos(\theta)) d\omega d\lambda}$$

spectral heat rate/heat flux

$$dq_{\lambda} = \frac{dq}{d\lambda} = I_{\lambda, e}(\lambda, \theta, \phi) dA_s \cos(\theta) d\omega$$

$$dq_{\lambda}'' = I_{\lambda, e}(\lambda, \theta, \phi) \cos(\theta) d\omega$$

$$= I_{\lambda, e}(\lambda, \theta, \phi) \cos(\theta) \sin(\theta) d\theta d\phi$$

Emissive Power, E (for diffuse surf, emission is isotropic)

$$E_{\lambda}(\lambda) = q''_{\lambda}(\lambda) = \int \int I_{\lambda, e}(\lambda, \theta, \phi) \cos(\theta) \sin(\theta) d\theta d\phi$$

$$E_{\lambda}(\lambda) = \pi I_{\lambda, e}(\lambda) \quad \text{intensity of emissive power}$$

$$E = \pi I_e \quad \text{for all wave lengths}$$

units ($\text{sr} \cdot \text{W/m}^2 \text{sr}$)

Irradiation, G (if radiation is diffuse)

$$G_{\lambda}(\lambda) = \pi I_{\lambda, i}(\lambda)$$

$$G = \pi I_i$$

Radiosity, J (of opaque surface)

$$J_{\lambda}(\lambda) = \pi I_{\lambda, p+\epsilon}(\lambda)$$

$$J = \pi I_{p+\epsilon} \quad (p+\epsilon = \text{reflected} + \text{emitted})$$

